Low-Level Windshear Detection from Passively Received Mode S EHS at Helsinki-Vantaa

Development phases, algorithm changes, their reasons, and the database checks of the EFHK study

Otso Laakso (OH2GAX)
Tuusula, Finland · github.com/oh2gax/mode_s_wind

Abstract. MODE-S Wind is a single passive 1 090 MHz receiver near Helsinki-Vantaa airport (EFHK) that decodes the Mode S Enhanced Surveillance (EHS) replies of aircraft to the ground interrogators. On final approach and on the climb-out every equipped aircraft reports its true airspeed (TAS) and ground speed (GS) about every two seconds; their difference is the headwind the aircraft is flying into. This log follows an attempt to turn that stream into an objective, per-flight report of low-level windshear — something that, to our knowledge, has not been published for passively received Mode S data. The study started with six simple browser algorithms (May – October 2026). A replay of 3 084 landings showed that they mostly reacted to the normal change of the wind with height, and on 24 October 2026 they were abandoned in favour of the headwind-ramp method of the Hong Kong Observatory (HKO), scaled by the length of the ramp, together with a comparison of each flight's 200-ft band winds with the flights of the previous 45 minutes. The log then records each later change — a relative rule for weak ramp advisories (26 October), a label that tells shear seen by several aircraft from single-aircraft turbulence (27 October), and a flight-phase correction of the band comparison (27 October) — with the evidence that led to it and its replay on the stored data.

Research only. All results come from a passive, uncertified receiver and describe air that an aircraft has *already* flown through. They are not an operational windshear warning and must not be used for flight operations or air traffic control.

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About this document

This is a *living* study log. It is written for aviation meteorologists and atmospheric scientists who want to follow how the method develops, and it is updated whenever the detection changes or a new check of the database has been made. The rules for keeping it are:

- Each change of the method gets its own phase section: what was changed, *why*, the evidence behind it, and the replay of the stored flights with the new version. Earlier sections are not rewritten; a later correction is marked as such.
- Every check of the database is added to the dated log in Section 10, also when nothing was changed.
- The current parameters are always in Section 11, the open questions in Section 12.

Times are UTC. Heights are feet above mean sea level (MSL) unless said otherwise; the EFHK thresholds are at about 180 ft. "kt per 1 km" means the 1-km equivalent of a headwind ramp (Section 3.2). The user-facing description of the present detector is the *Windshear User Guide* [22]; the retired algorithms are documented in [21].

Table 1. Version history of this document.

Version	Date	Content
1.0	27 Oct 2026	First edition: Phases 0–4 (May – 27 October 2026), the database checks of 24–27 October 2026, parameters and open questions.
1.1	27 Oct 2026, evening	Phase 5: higher band limits for turboprop departures (v6); the evening data check of 27 October; results and parameters updated to v6.
1.2	27 Oct 2026, late evening	New tools (Section 8): the space–time diagram of the band changes, the raw climb-out series and the trial ramps along the departures; next steps updated.

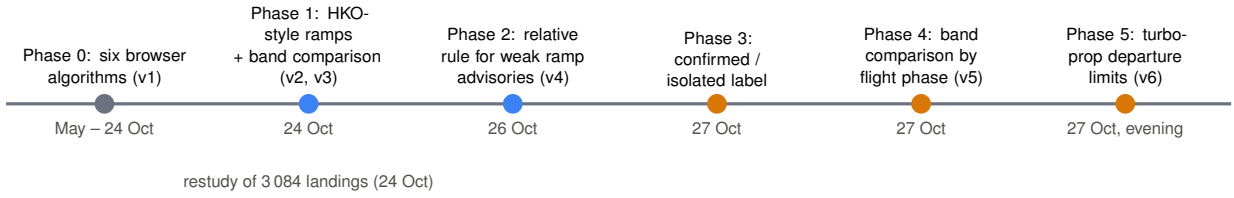


Figure 1. The phases of the study (not to scale). The stored report version (v) is given in brackets.

1 Setting and data

1.1 Receiver and airport

The receiver (a Jetvision Radarcape with a ground-plane antenna) is in Tuusula, north of EFHK, and feeds a Raspberry Pi 4 that decodes, stores and analyses the data. EFHK has three runways: 04L/22R, 04R/22L and 15/33. The Finnish surveillance network — Mode S radars and the wide area multilateration (WAM) system [14, 15] — interrogates the aircraft; MODE-S Wind only listens to the replies.

1.2 The Mode S EHS registers used

The Comm-B replies (DF 20/21) carry the requested register, without its identifier [3]. Two registers matter here:

- **BDS 5,0, track and turn report**: roll angle, true track, ground speed, track angle rate and **true airspeed**. GS and TAS have 2-kt resolution.
- **BDS 6,0, heading and speed report**: magnetic heading, indicated airspeed, Mach number and two vertical rates.

The wind vector of an aircraft follows from a BDS 5,0 / 6,0 pair (ground vector minus air vector, with a heading calibration learnt per airframe) [12]. For windshear the more useful quantity is simpler: **TAS − GS from one BDS 5,0 reply**. It needs no heading, both speeds come from the same moment, and it equals the along-track headwind plus a small, slowly varying crab term $\mathrm{TAS}(1-\cos\beta)$ (about 2 kt at a 10° crab angle β), which cancels in changes along one flight.

1.3 What is stored

- Every observation (position, altitude, decoded registers) since 24 September 2026.
- **Raw approach series** since 9 October 2026: for every landing the BDS 5,0 and 6,0 replies and the 3-s position sweeps from about 12 NM to the lowest point seen. Since 24 October also the ADS-B velocity messages (ground speed at about 2 Hz).

- **Band profiles** of every landing and, since 11 October, every departure: the mean wind vector, TAS − GS and the number of wind pairs in 200-ft bands.
- **A windshear report** (`shear_json`, about 0.35 kB) for every landing and departure since 24 October 2026; older flights were back-filled from the stored data. Every change of the method raises the report version, and the back-fill then recomputes all stored flights once, so the counts in this log always refer to one method over the whole period.
- For comparison: the MEPS model of MET Norway [13] (optional; nothing in the detection depends on it), the Jokioinen soundings, and the EFHK METARs.

Table 2. Data quality on final approach, 3 084 landings 9–23 October 2026 [16].

Quantity	Result
BDS 5,0 reply interval	median 1.97 s (p90 3.95 s), about 144 m along the path
Interval between changes of TAS − GS	median 3.96 s (about 290 m, 50 ft on a 3° path); 23 % of the replies repeat the previous register
Per-reply headwind noise (RMS)	median 1.06 kt (p90 1.50); the 2-kt quantisation alone gives 0.82 kt
Lowest reply above the threshold	15: 232 ft · 22L: 337 · 22R: 349 · 04L: 508 · 04R: 733 ft (median by runway; set by the receiver's line of sight)
Largest 1-km headwind loss per approach	median 4 kt, p99 8 kt (smoothed profile, benign weather)
Horizontal F-factor over 1 km	median 0.014, p99 0.03, maximum 0.065 (threshold 0.105 never reached)

A 15-kt change over 0.4–4 km is therefore about ten times the noise of the difference of two 400-m means. The practical floor is set by the natural variability of the wind, not by the data.

2 Phase 0 — six browser algorithms (May – 24 October 2026) and why they were abandoned

2.1 What they were

The first windshear view ran in the browser of each viewer, only while the page was open. It polled the latest heading-calibrated wind of each aircraft in the approach corridor every 3 s, turned it into a headwind on the runway heading and applied one of six algorithms, chosen per browser. Levels were Monitor 10 kt, Warning 15 kt and Alarm 25 kt (loss or gain), with a two-poll confidence gate. Full details are in [21].

Table 3. The six v1 algorithms, their main problem, and the share of landings that would have reached Warning ($\geq$ 15 kt) in 14 benign days (9–23 October 2026; highest METAR wind 17 kt, no windshear reported) [16].

Algorithm	Compared	Main problem	Warning
Energy	GS + alt/100 over 45 s	ground speed instead of airspeed; 1 kt equals about 12 ft at approach speed, not 100 ft; the planned descent and deceleration look like an energy loss	97.9 %
Gradient	largest headwind difference of two points 200–3 000 ft apart	spans up to 17 km of path; the maximum over all pairs grows with noise; no smoothing, no length scaling	13.3 %
Pairwise	two aircraft 200–2 000 ft apart	measures the normal wind profile plus two airframes' own TAS and heading biases	7.6 % of pairs
Rate	current headwind vs older stored points	look-back up to 2 000 ft / 3 NM without length scaling	3.8 %
Kinematic	IAS − GS over 45 s	IAS is not TAS (about 4 kt at 2 000 ft falling to 1 kt at 500 ft), so the descent itself fakes a headwind change	2.3 %
Baseline	headwind vs the mean wind below 2 000 ft of the last 30 min	mixes all heights into one expected value	0.2 %

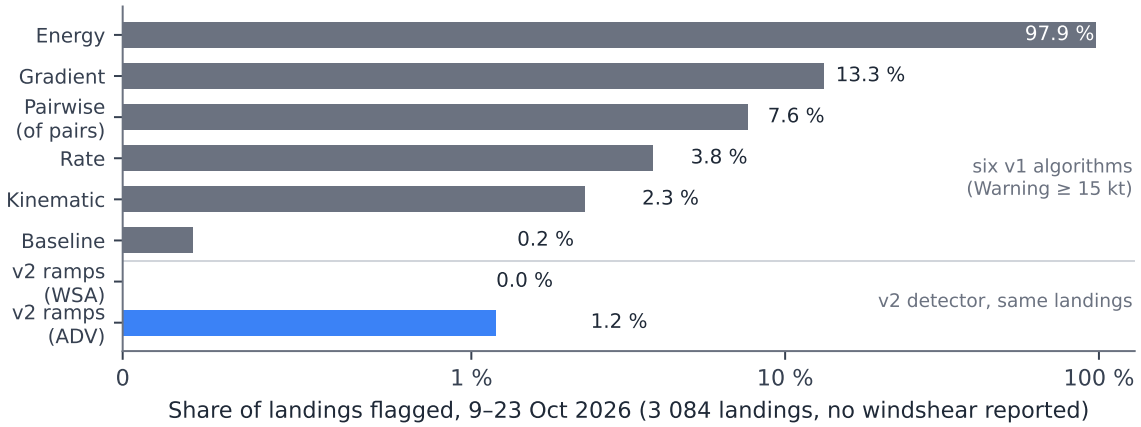


Figure 2. Share of landings flagged in 14 benign days by each v1 algorithm (Warning, grey) and by the v2 ramp detector on the same landings (blue: advisory, orange: windshear alert). Logarithmic axis.

2.2 The common faults

- **No scaling with the distance of the change.** A 15-kt change means very different things over 400 m and over 10 km. Every operational method — the HKO glide-path ramps, the 1-km F-factor, the station spacing of LLWAS — ties the threshold to a length or a time [4, 8, 10].
- **Altitude instead of distance along the path**, which mixes shear along the path with the ordinary increase of the wind with height — the "gentle ramps" that HKO filters out [6].
- **Loss and gain treated alike**, although the headwind loss is the performance-decreasing case, and an Alarm level (25 kt) that matches no standard.
- **One poll's wind per aircraft**, noisier than the TAS − GS of the replay.

Kept from v1: the gating of the corridor and of an established final, the go-around detection and the guard against frozen GNSS positions. The decoded BDS 4,5 hazard flags were also examined: 47 % of them showed a windshear level above 0 in a period with no windshear, so they are other registers misidentified as BDS 4,5 and must not feed any alert.

2.3 The decision: one method after the Hong Kong Observatory

On 24 October 2026 all six algorithms were retired at once (no parallel "legacy" mode). The replacement follows the GLYGA method of the Hong Kong Observatory, which finds headwind *ramps* along the glide path in Doppler-lidar scans and alerts when $\Delta V / L^{1/3}$ exceeds a threshold for ramp lengths L of 400–4 000 m [4, 5, 7]. The reason for choosing it is that the lidar's input — a one-dimensional headwind profile along the approach path — is exactly what each approaching aircraft samples with its own BDS 5,0 replies. The $L^{1/3}$ scaling comes from flight-data studies of how an aircraft responds to a change of headwind, so it carries aircraft-performance meaning that the v1 thresholds lacked. HKO has also noted that Mode S downlinked winds are, in principle, fast enough for real-time windshear alerting [16], but no evaluation has been published; this study tries to provide one.

3 Phase 1 — detector v2 (24 October 2026)

The detection moved to the server. It runs for every landing and departure whether or not anyone has the page open, and each flight gets a stored report.

3.1 Headwind profile along the path

For each reply TAS − GS is computed. Replies are dropped when the roll exceeds 5°, when the register repeats (same GS and TAS within 6 s), or when they are more than 12 kt from the running median of ±7.5 s. The distance along the path comes from the integrated ground speed. The profile is a 400-m running mean on a 100-m grid.

3.2 Headwind ramps

For every grid point and length $L \in \{400, 600, 800, 1\,000, 1\,500, 2\,000, 3\,000, 4\,000\}$ m the change ΔV over L is scaled to a 1-km equivalent

$$\Delta V_{1\mathrm{km}} = \Delta V \left(\frac{1\,\mathrm{km}}{L} \right)^{1/3}, \tag{1}$$

and the strongest loss and the strongest gain of the flight are kept. A ramp counts only if at least 60 % of it is covered by data and $|\Delta V|$ is at least five times its own noise; ramps of 600 m or less need three replies at each end and must persist over the next 400 m. Only ramps in the low-level zone count: below 1 600 ft above the threshold or within 5 NM of it [1, 2]. The levels, in kt per 1 km:

ADV advisory ≥ 10 **WSA** windshear alert ≥ 15 **MBA** microburst alert ≥ 30 (loss only)

Beside the ramps, the horizontal F-factor $F_h = \dot{W}_x/g$ is computed as a mean over 1 km [8, 9]; the vertical term $-w/V$ cannot be observed from Mode S.

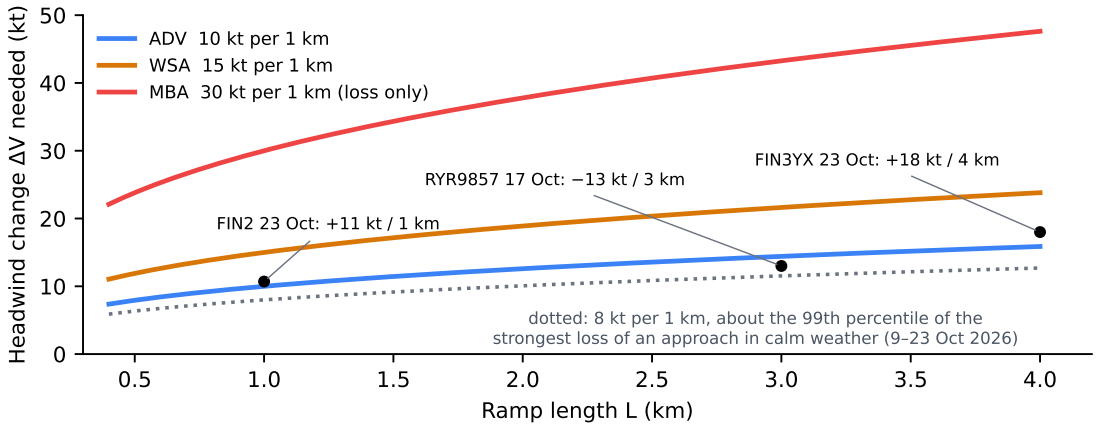


Figure 3. The headwind change needed for each level as a function of ramp length. The dotted line is about the 99th percentile of the strongest loss of an approach in calm weather.

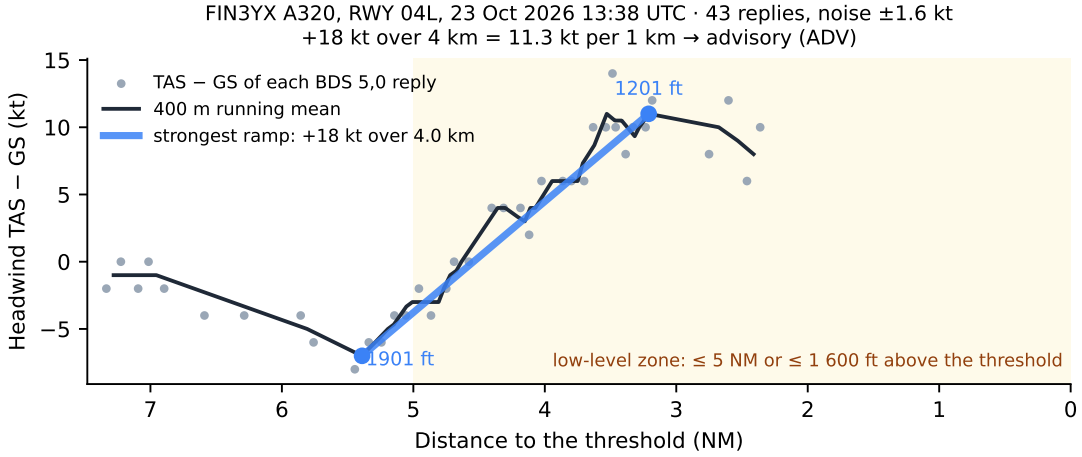


Figure 4. A ramp in practice: the per-reply headwind, the 400-m mean and the strongest ramp of one landing (23 October 2026, a persistent shear layer before a front).

3.3 Band comparison between flights (v2 → v3 the same day)

A ramp shows what one aircraft met. A front or gust front shows as a change *between* flights. For this the wind of each 200-ft band from 400 to 1 800 ft is compared with the median of the flights of the previous 45 minutes (at least 3 references); a change of ≥ 10 kt is ADV and ≥ 15 kt WSA.

The first version (v2, morning of 24 October) compared the band means of TAS − GS with *landings on the same runway heading*. A check the same afternoon [17] showed that this did not work for departures: 55 % of departures had no landing on the same runway direction in the previous 45 minutes (the usual configuration is landing 15, departing 22R), and climb-outs give too few replies within ±15° of the axis.

Why the change. TAS − GS is tied to the flight's own direction, so it can only be compared with flights in the same direction. The **wind vector** of each band (from the BDS 5,0 / 6,0 pairs) can be compared across all runways. In v3 (evening of 24 October) the reference became the median u, v of all flights of the last 45 minutes — landings and departures — and the difference is projected on the flight's own direction:

$$\Delta H_b = -\left(\mathbf{w}_b - \mathrm{med}_{\mathrm{ref}}\,\mathbf{w}_b\right)\cdot\hat{\mathbf{d}}, \tag{2}$$

where $\hat{\mathbf{d}}$ is the unit vector of the flight direction and a negative ΔH_b is a headwind loss. Bands with corrupt values (speed above 80 kt or spread above 15 kt) are skipped. Adding departures raised the share of landings with at least 3 references from 88.9 % to 92.5 % and filled the night gaps when there is about one landing per hour.

3.4 First results

Table 4. v2/v3 replay on the stored flights (benign weather apart from 17 October).

	Flights	ADV	WSA
Landings 9–23 Oct, ramps	3 084	37 (1.2 %)	0
Landings 9–24 Oct (v3), ramps / band	3 310	38 / 0	0 / 4
Departures 11–24 Oct (v3), band	3 084	21 (0.7 %)	9 (0.3 %)

Most ramp advisories were real low-level wind-profile cases, such as the night inversion of 11 October (+17 kt over 3 km between 1 800 and 1 200 ft in successive landings) and the persistent shear layer before the front of 23 October. The band WSAs were the wind change of **17 October 2026** on runway 22L (Figure 5): FIN2EZ, RYR9857 and FIN2AT at 09:04–09:07 had 25–29 kt less headwind at 800–1 200 ft than the landings ten minutes earlier. The departure FIN863 from 22R had already shown −24.7 kt at 1 200 ft at 08:54, *ten minutes before the first landing*; the departures that switched to runway 04 at 09:13–09:38 then saw the wind reversal (+20…+38 kt). The fourth band WSA was on 18 October at 08:41, and a departure on 24 October at 04:42 showed +16 kt at 1 800 ft during the night frontal passage. The noise of the reply stream stayed at a median of 1.24 kt.

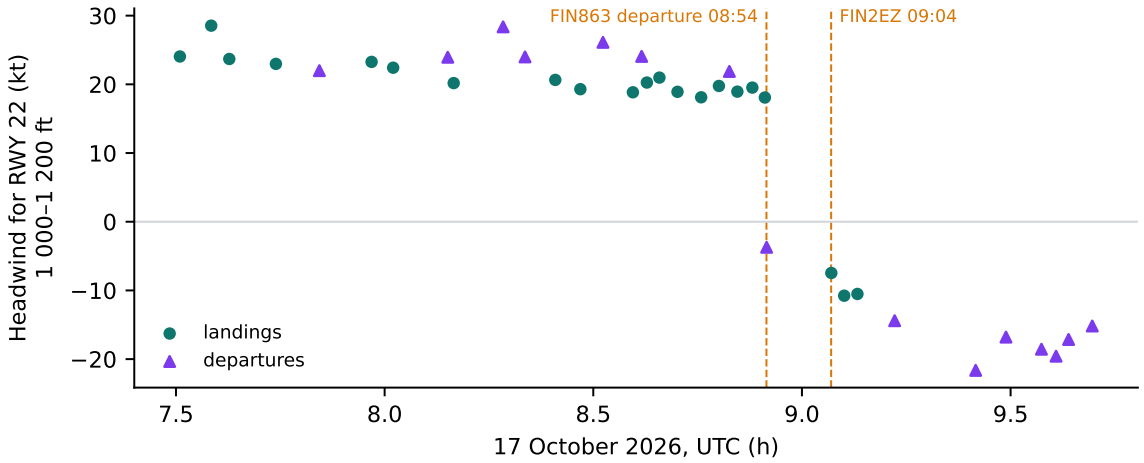


Figure 5. The 17 October 2026 case: band headwinds of the landings on 22L before and after the wind change.

4 Phase 2 — relative rule for weak ramp advisories, v4 (26 October 2026)

4.1 Observation

On the evening of 25 October a stable boundary layer formed under a strong southerly flow: the median headwind of the landings rose from 20 kt at 400 ft to 35 kt at 1 800 ft. Between 20:00 and 22:30 UTC 10 of 39 landings got a ramp advisory. Each ramp was real — the aircraft did lose that headwind — but every aircraft lost the same amount at the same heights, and the band comparison saw nothing (largest change 7.6 kt). The advisory flagged the expected profile, not anything unusual. In the whole period 9–26 October, 59 landings (1.6 %) had a ramp advisory, all at 10–14 kt per 1 km [18].

4.2 Change

A ramp of 10–12 kt per 1 km stays an advisory only if it exceeds the change that the reference flights had over the same heights by at least 10 kt per 1 km. With $H_{\mathrm{ref}}(a)$ the median reference headwind at altitude a (from the band references of Section 3.3) and a_1, a_2 the ends of the ramp:

$$e = H_{\mathrm{ref}}(a_2) - H_{\mathrm{ref}}(a_1), \qquad r = \frac{|\Delta V - e|}{|\Delta V|}\,\Delta V_{1\mathrm{km}} \quad (\text{when } \Delta V - e \text{ has the sign of } \Delta V, \text{ else } r = 0), \tag{3}$$

and the ramp is ADV if $\Delta V_{1\mathrm{km}} \geq 12$ or $r \geq 10$. Without references (at night) the absolute rule applies. **WSA and MBA stay absolute.**

> **Why the change.** The advisory is meant to point at something the other aircraft did *not* have. The ordinary increase of the wind with height in a stable layer is a known, "gentle ramp" situation that HKO also removes from its alerts [6]. The stronger levels are kept absolute because a 15-kt-per-km loss matters to the aircraft whether or not the previous aircraft had it too.

Replay 9–26 October: landing advisories 59 → 26; in the windy days 24–26 October 21 → 13. The band WSAs (4) and the departures were unchanged.

5 Phase 3 — confirmed or isolated (27 October 2026)

5.1 Observation: the turbulent southerly day of 26 October

26 October brought a strong, warm and moist southerly flow: surface wind 12–17 kt (gusts to 28–33 kt at 02–03 UTC), 30–40 kt at 1 000–4 000 ft, and a near-neutral lapse rate in the Jokioinen soundings — mechanical turbulence. The Approach Conditions Index of the landings had its highest median so far (4.2; typically 1.1–2.2) and the per-reply noise rose from about 1.2 to 1.86 kt [19]. Of 200 landings in 24 hours, 12 had a ramp alert (9 ADV, 3 WSA). Looking at each case with its neighbours:

- **Real shear layer:** FIN6KB at 13:08 on 22L. Every 22L arrival lost 7–10 kt of headwind at 1 150–1 350 ft, and four neighbours had a matching ramp.
- **Partly confirmed:** FIN7SM at 14:06 (WSA): an 11-kt ramp over 400 m at about 1 100 ft; FIN7RL had the same dip at 1 000 ft three minutes earlier — a real, short eddy or gust.
- **Isolated:** seven landings (two of them WSA) had a short ramp that no neighbour within ±15 minutes repeated, while their 200-ft band profiles looked like their neighbours'. These are turbulence rather than windshear.

5.2 Change

A ramp alert is labelled **confirmed** when another landing on the same runway direction within ±15 minutes had a ramp of the same kind (loss or gain) of at least 7 kt per 1 km whose middle lies within ±300 ft of this one's; otherwise **isolated**. The label is first computed from the landings before and completed 15 minutes later from the landings after. *The alert level is not changed.*

> **Why the change.** A shear layer, front or gust front is met by several aircraft; a turbulent eddy over rough terrain by one. The label lets the reader — and later statistics — separate the two. The level is kept because the first aircraft into a new shear is always alone at that moment; the 17 October departure FIN863 is the example.

Result on the stored data (9–27 October): of 36 landings with a ramp alert, 25 are confirmed and 11 isolated; 10 of the 11 isolated ones are from the turbulent day of 26 October (Figure 9).

6 Phase 4 — band comparison by flight phase, v5 (27 October 2026)

6.1 Observation: departures read more headwind than landings

Already in the first version a constant offset had been seen: in the same bands, departure TAS − GS was 2.3 kt above that of landings (1 946 replies, 9–23 October), and v2 simply subtracted it. When the band comparison moved to the wind vector (v3) the correction was dropped, but the offset remained: departures read +1.4…+1.7 kt more headwind than the landings within ±20 minutes (4 543 band observations), and in the windy night of 26–27 October +4…+9 kt. Of the 36 departure band flags up to 27 October, 27 were "more headwind" and only 9 "less".

6.2 Tests

- **Acceleration (a lag between the GS and TAS sources):** not the cause. Among the departure observations the offset hardly depends on the acceleration (slope -0.64 kt per kt s^{-1}, $r = -0.10$), and in the raw landing series the *steady* segments have the largest offset (-1.5 kt) while decelerating ones have about 0.
- **Geography (departures and landings fly through different air):** not the cause. Departures to 04 read +1.7 kt and to 22 +1.4 kt.
- **Comparison with the MEPS model** at 600–1 800 ft (model levels, pressure from QNH), headwind along each flight's own direction: landings -1.45 kt, departures +1.38 kt (Table 5, Figure 6). The offset is *symmetric about the model*, and the wind speeds of departures agree with the model exactly.

Table 5. Measured minus MEPS area-mean wind, 600–1 800 ft, 9–27 October 2026.

Flights	Band means	Headwind − MEPS	Wind speed − MEPS
Landings	9 439	−1.45 kt	−0.9 kt
Departures	6 789	+1.38 kt	0.0 kt

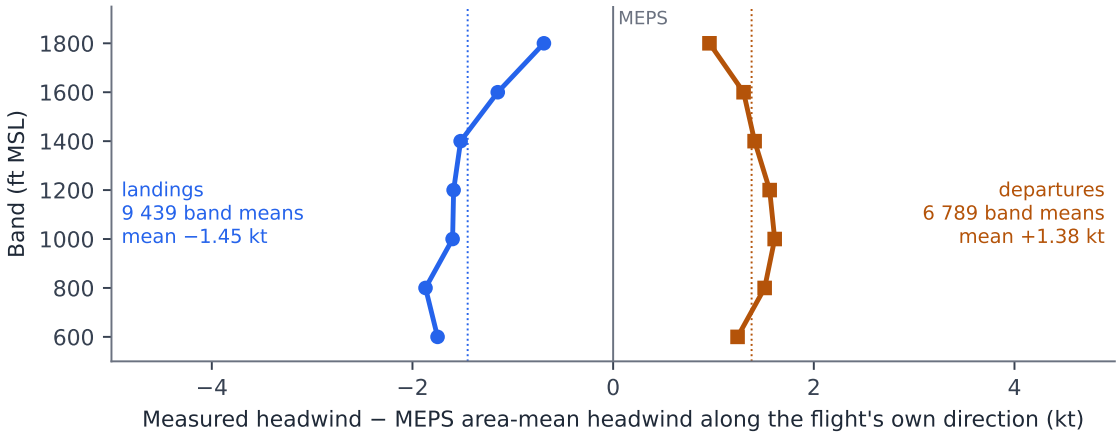


Figure 6. Headwind of landings and departures minus the MEPS area-mean headwind along each flight's own direction, by 200-ft band. The dotted lines are the means used as the phase correction.

The most likely explanation is an airspeed effect of the flight phase — the reported TAS of an aircraft in approach configuration (flaps and gear out, low speed, high angle of attack) is slightly low and that of a climbing aircraft slightly high, a pressure-error / calibration effect of the air-data system — rather than a real wind difference. It does not matter for the ramps, because the offset is constant within one flight and cancels in ΔV. It does matter for the band comparison, which mixes landings and departures.

6.3 Change

A band is compared with **references of the same phase** (landings with landings, departures with departures) when it has at least 5 of them. Otherwise all references are used, but every vector — the references' and the flight's own — is first corrected by the phase bias b along its own direction:

$$\mathbf{w}' = \mathbf{w} + b_{\text{phase}}\,\hat{\mathbf{d}}, \qquad b_{\text{arr}} = -1.45\,kt, \quad b_{\text{dep}} = +1.38\,kt. \tag{4}$$

The report records which was used (`ph`: same / all). A first trial with 3 same-phase references produced a spurious landing WSA on 17 October at 09:46, so 5 was chosen.

Why the change. At night there are often only a few landings, and departures are the only recent references; mixing them is valuable, but only after the systematic phase difference has been removed. Same-phase references need no correction at all, so they are preferred whenever there are enough of them.

Table 6. Replay of v4 and v5 on all stored flights, 9–27 October 2026.

	v4	v5
Landings: band WSA	4	4
Landings: mean band change	−0.79 kt	+0.21 kt
Departures: mean band change	+1.57 kt	−0.33 kt
Departures: band ADV + WSA	27 + 9	28 + 7
Departures: flags with more / less headwind	27 / 9	19 / 16
Departure ADVs 26–27 October	5	2
17 October 09:34 departure	WSA 20.2 kt	ADV 12.2 kt

The departure flags are now balanced between more and less headwind, as they should be if they come from the weather and not from the method.

7 Phase 5 — higher band limits for turboprop departures, v6 (27 October 2026, evening)

7.1 Observation: ATR departures flag five times as often

In the evening data check of 27 October [20] all three departure advisories of the day were ATR 72 climb-outs (FIN4BX, FIN2PE, FIN6WR), each with 11–15 kt *less* headwind at 1 600–1 800 ft. Over all stored departures, 20 of 688 turboprop climb-outs (2.9 %) had a band advisory or alert, against 18 of 2 946 jet climb-outs (0.6 %); the turboprop flags were mostly "less headwind" (32 of 49 changes of at least 8 kt), and largest at 1 400–1 600 ft. The landings of the same turboprops showed nothing unusual (0.3 %).

7.2 Tests

The question was whether the cause is an *offset* of the aircraft type (to be subtracted) or a larger *scatter* (to be allowed for). The band comparison was replayed for every flight of 13–27 October with the change of *every* band, not only the strongest one (the strongest band of each flight gives a biased picture, because it is selected for its size).

- **Per-type offset:** real but small, and repeatable. Learnt on 13–18 October and compared with 19–27 October, the type means agree ($r = 0.90$ over 33 type–phase pairs): E190 +1.3 / +1.4 kt (departures / landings), A220 +1.3 / +1.2, A330-300 −0.7 / −1.4, A321 −0.3 / −1.1, ATR 72 −0.9 / −0.2 kt. Removing them (test period) narrowed the scatter of the jets by about a tenth (landings 2.12 → 1.92 kt, departures 2.64 → 2.51 kt), but hardly that of the turboprop departures (3.26 → 3.23 kt); their flags fell only from 3.3 to 2.5 %.
- **Scatter:** the turboprop climb-outs scatter more (Figure 7) — SD 3.2–3.8 kt at 1 200–1 800 ft against 2.7–2.9 kt for the jet departures — although they have three to seven wind pairs per band (they climb slowly). The heights are where an ATR accelerates, retracts its flaps and makes its first turn; the band means there are also 1.0–1.6 kt low. The ratio of the scatters is about 1.3.

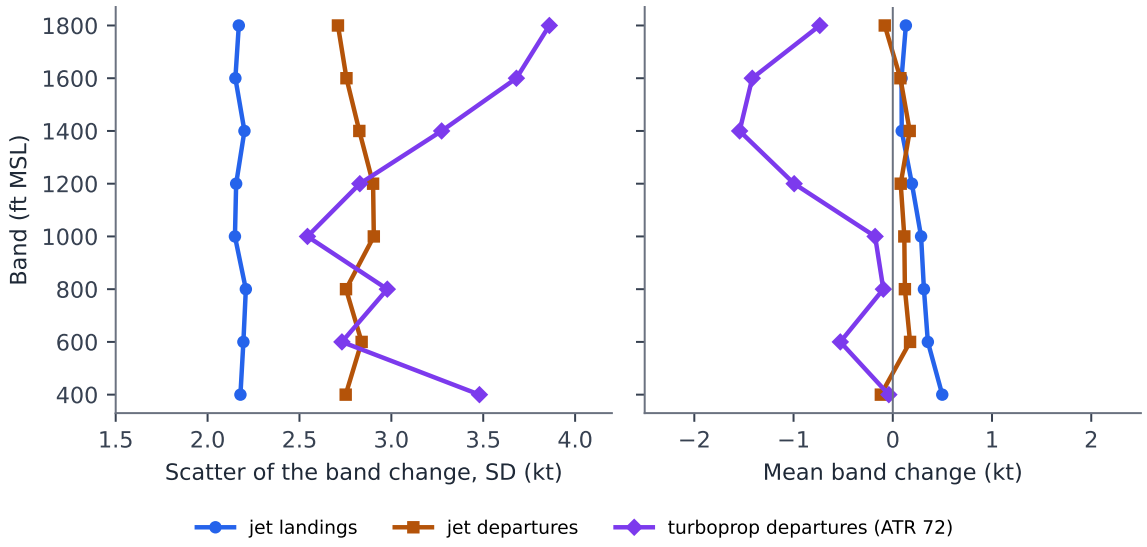


Figure 7. Band change of each flight against the flights of the previous 45 minutes, every band, 13–27 October 2026 (v5): scatter (left) and mean (right) for jet landings (20 136 band changes), jet departures (14 713) and turboprop departures (2 879, almost all ATR 72).

7.3 Change

For the departure of a turboprop (ICAO types ATR 42/72, Dash 8, Saab 340/2000, Fokker 50, Do 328, Jetstream, Beech 1900 / King Air, PC-12, Caravan, L 410, Brasilia, Metro) the band limits are multiplied by `TP_DEP_SCALE` = 1.3: **advisory 13 kt, windshear alert 19.5 kt**, also in the rule for a band resting on one wind pair. Jets and all landings keep 10 / 15 kt. The report records the limit used (`thr`), and the hover of the alert and of the ACI + WS chart shows it.

Why the change. A limit should stand at the same distance from the normal variation for every aircraft. The jets' 10 kt is about 3.5–4 times the scatter of their climb-outs; 13 kt keeps the ATR at the same distance. The per-type offset is kept in reserve: it is too small to matter for the alerts, but may be worth removing later for the accuracy of the band winds themselves.

Table 7. Replay of v5 and v6 on all stored flights, 9 / 11 – 27 October 2026 18 UTC.

	v5	v6
Turboprop departures (711): band ADV + WSA	18 + 3	8 + 1
Jet departures (2 987): band ADV + WSA	13 + 4	13 + 4
Landings (3 913): all levels	—	unchanged
27 October: FIN6WR (−10.7 kt) / FIN4BX (−14.2) / FIN2PE (−14.9)	ADV / ADV / ADV	none / ADV / ADV

Of the 14 turboprop departures whose level changed, 12 went from ADV to none (changes of 10.1–11.9 kt) and 2 from WSA to ADV (16.5 and 19.3 kt). The departures now flag at a similar rate whatever the type: 1.3 % of the turboprops, 0.6 % of the jets.

8 New tools for the study (27 October 2026, late evening)

Two additions that change nothing in the alerts but give the study new material.

8.1 Space–time diagram of the band changes

The detection stores for each flight only its *largest* band change. To see how a change travels through successive flights, the change of *every* band of every landing and departure is now recomputed on request from the stored band profiles, exactly as in the band comparison (same references, phase correction and sign), and drawn as a time–height picture: the **ST** view of the Windshear page. Each cell is the median of the flights in a 5–30-minute slot. A front or a shear layer is a coloured streak through successive flights; the eddy of a single aircraft is damped by the others in its slot. Figure 8 shows the wind change of 17 October: the whole layer from 600 to 1 800 ft lost headwind at 09:00 (the landings FIN2EZ, RYR9857, FIN2AT and the departure FIN863 before them), and the next slots show the reversal after the runway change. On the turbulent 27 October the same diagram has only scattered single cells.

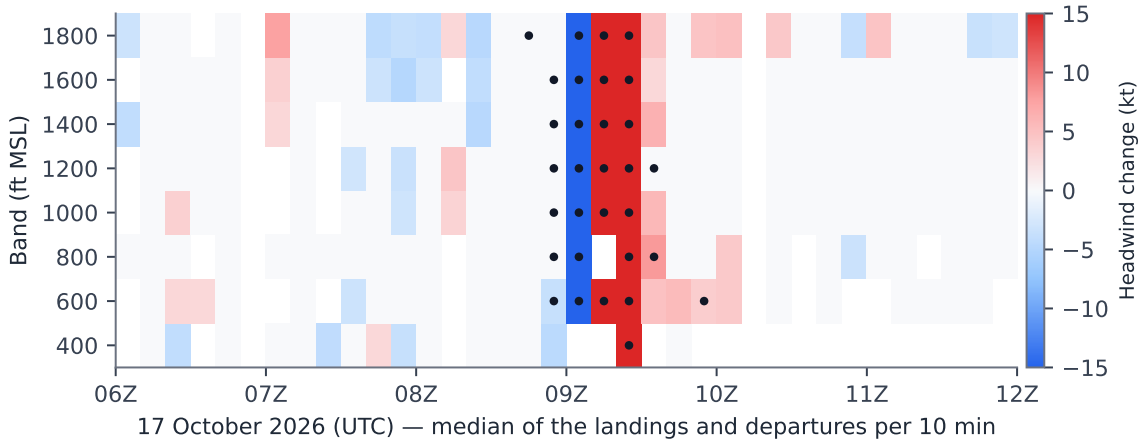


Figure 8. Space–time diagram, 17 October 2026 06–12 UTC: the median headwind change per 10 minutes and 200-ft band of the landings and departures, each against the flights of the previous 45 minutes (blue less, red more headwind; changes below 3 kt white). Dots: a flight at or beyond its advisory limit.

8.2 Climb-out series and trial ramps along the departures

Until now the departures stored only their band profile, so the ramp analysis could not be applied to them. From 27 October every climb-out keeps its reply series (BDS 5,0 / 6,0 replies with the height at the reply time, the ADS-B velocity messages and the sweeps; table `departure_series`, the format of the approach series), and at the end of the climb the ramp analysis of the landings is run on it — the low-level zone being the first 1 600 ft above the airport or 5 NM from it. The result is stored in the departure report as `trial`, with the level it *would* have; **no alert is raised**. Before the ramps may alert, the stored climb-outs must show how the acceleration (speed changes and flap retraction in the first 1 500 ft) affects TAS − GS, and whether the 15–25 replies of a fast climb in the zone (about 100 on final) are enough. Planned check: after one to two weeks, compare the trial ramps of successive departures and with the landings' ramps and band changes.

9 Results so far

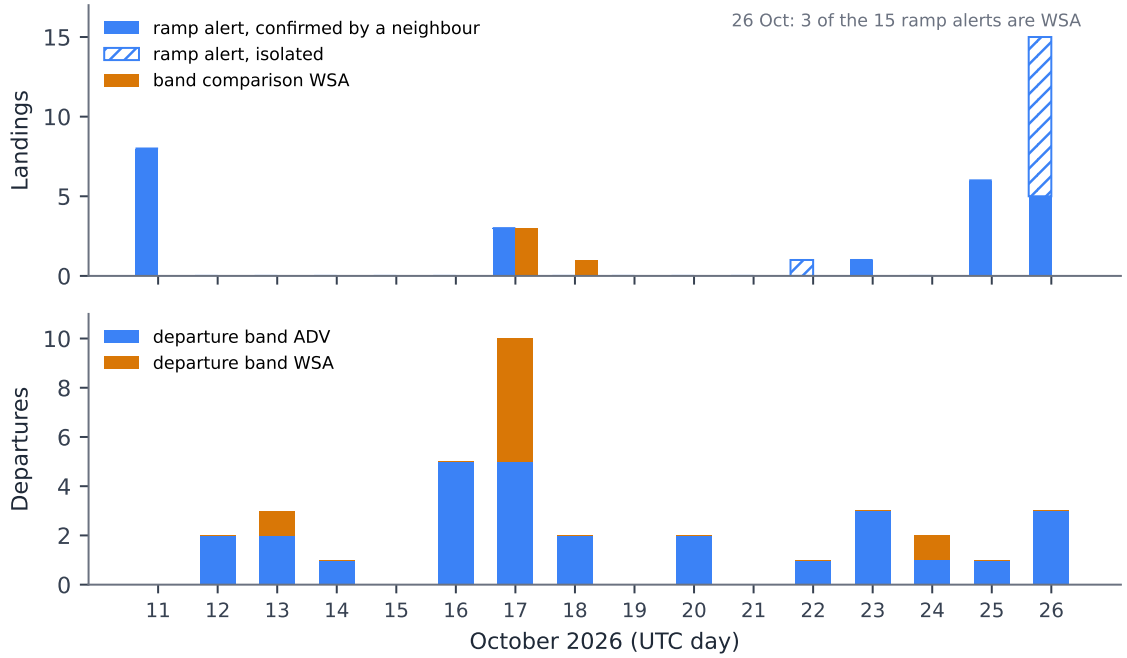


Figure 9. Windshear flags per UTC day with the method v5, 11–26 October 2026 (departure reports start on 11 October). Top: landings, ramp alerts split into confirmed and isolated, and band-comparison WSAs. Bottom: departure band flags.

Table 8. All stored flights with the current method (v6), 9 October – 27 October 2026 18 UTC.

	Flights	ADV	WSA	MBA
Landings, ramps	3 913	41 (1.05 %)	3 (0.08 %)	0
of which confirmed / isolated			29 / 14 (+1 open)	
Landings, band comparison	3 913	1 (0.03 %)	4 (0.10 %)	—
Departures, band comparison	3 698	21 (0.57 %)	5 (0.14 %)	—

The period so far contains no convective weather, and no windshear was reported by METAR in it, so the numbers above describe the method in autumn boundary-layer weather: stable nights, a frontal passage (23–24 October), one sharp wind change (17 October) and two turbulent, windy days (26–27 October).

10 Database checks

Every check is made on a read-only copy of the station database (a "snapshot"), with the logs and the latest soundings. Newest last.

Date	Check and main results
24 Oct 2026	**Restudy of the v1 algorithms** (snapshot 23 Oct; 3 084 landings 9–23 Oct). Noise floor, coverage and false-alarm replay (Tables 2 and 3). Literature review: no published windshear study on passively received Mode S EHS. BDS 4,5 hazard flags found unreliable. ⇒ v1 retired, v2 designed.
24 Oct 2026	**Post-front check** (snapshot 24 Oct 16 UTC). 350 landings: 7 ramp advisories, all gains of 10–11 kt per km on 04L/R on 23 Oct 11:34–14:15 (a persistent shear layer at 1 400–1 800 ft, seen by the other landings at 6–9 kt). Runway changes (04 → 15 → 22) gave no false alarms. The TAS − GS band comparison could not compare 55 % of departures. ⇒ v3 (vector bands, departures).
25 Oct 2026	**Post-deployment check** (night of 24–25 Oct). 98 landings, 109 departures, all levels none; F_h median 0.012, p99 0.026; noise 1.24 kt. ADS-B ground speed collected in the approach series: median 376 messages vs 100 BDS 5,0 replies per landing; BDS 5,0 GS minus ADS-B GS +0.47 ± 0.76 kt.
26 Oct 2026	**Data check** (snapshot 26 Oct 14 UTC). 59 ramp advisories 9–26 Oct (1.6 %), 10 of 39 landings on the stable evening of 25 Oct. ACI rises monotonically with the wind (median 1.1 below 8 kt, 6.1 at 18 kt and gusts). ⇒ v4.
27 Oct 2026	**Data check** (snapshot 27 Oct 05 UTC). The turbulent southerly day: 12 landing ramp alerts in 24 h, one real shear layer (FIN6KB), several isolated eddies. Departures read about +2 kt more headwind than landings over ∼11 000 band comparisons. MEPS check at the aircraft positions and the soundings normal. ⇒ confirmation label.
27 Oct 2026	**Phase-offset analysis** (same snapshot). Acceleration and geography ruled out; the offset is symmetric about MEPS (−1.45 / +1.38 kt). ⇒ v5; replay in Table 6.
27 Oct 2026, evening	**Data check** (snapshot 27 Oct 18 UTC, the first downloaded gzip-compressed from the station). South-westerly flow with a low-level jet (34–40 kt at 0.75–1.1 km), ACI median 3.8. 156 landings since 02 UTC: 11 ADV, no WSA; ramp alerts 6 confirmed, 3 isolated, 1 open; one band ADV (JAL47, a B787 reading 207°/21 kt where its neighbours had 223–228°/26–32 kt — one aircraft). 167 departures: 3 ADV, all ATR 72. Soundings: 12 UTC T bias −0.03 K (RMS 0.44 K), the best launch so far. *Data quality:* on the station the magnetic declination model was missing from 12:54 to 18:25 UTC (fixed 10.5°; exact at EFHK, so no effect on the windshear analysis; up to 2° off far from the airport). ⇒ type and turboprop replay.
27 Oct 2026, evening	**Type-bias and turboprop replay** (same snapshot; every band, 13–27 October, offsets learnt on the first half and tested on the second). Per-type offsets small and repeatable; turboprop departures scatter 1.3 times as much as jet departures. ⇒ v6; replay in Table 7.

11 Current method and parameters (version v6)

Table 10. Parameters of the detection on 27 October 2026.

Item	Value
Headwind	TAS − GS of each BDS 5,0 reply; drop roll > 5°, repeated registers (< 6 s), > 12 kt from the ±7.5-s median
Profile	distance from integrated GS; 400-m running mean on a 100-m grid
Ramps	L = 400–4 000 m; $\Delta V\,(1\,\mathrm{km}/L)^{1/3}$; coverage ≥ 60 %; ≥ 5$\sigma$; short ramps (≤ 600 m): 3 replies per end, persist 400 m
Zone	below 1 600 ft above the threshold or within 5 NM
Ramp levels	ADV ≥ 10 (10–12 only if ≥ 10 beyond the reference profile), WSA ≥ 15, MBA ≥ 30 (loss) kt per 1 km
Confirmation	same runway direction, ±15 min, same kind, ≥ 7 kt per 1 km, middle within ±300 ft; label only
F-factor	horizontal term, mean over 1 km (reported, not alerted)
Band comparison	200-ft bands 400–1 800 ft; wind vector; median of the flights of the last 45 min (≥ 3); same phase if ≥ 5, else all with the phase bias removed (−1.45 / +1.38 kt); ADV ≥ 10, WSA ≥ 15 kt; a turboprop's departure ADV ≥ 13, WSA ≥ 19.5 kt (limits × 1.3)
Live operation	server-side every 3-s sweep; two analyses to raise an alert; an alert holds 10 min; 24-h log

12 Limits, open questions and next steps

12.1 Known limits

- **Coverage:** the last 0.6–2.3 NM before the threshold (below 230–730 ft, depending on the runway) is not seen from the receiver site. Microburst outflows are strongest there.
- **The vertical term** of the F-factor ($-w/V$, the downdraft) cannot be observed.

- **After the fact:** every report describes air an aircraft has already flown through; an event can fall between two landings (one every 2–3 minutes in busy hours).
- **Scales below about 500 m** are under the sampling scale of the replies; the reactive on-board warnings in QAR studies act at about this scale [16].

12.2 Next steps

1. **ADS-B ground speed** (about 31 October – 7 November 2026): replay the ramps with the ADS-B velocity messages, which arrive four to six times as often as the distinct BDS 5,0 values and follow the aircraft lower; the question is how much the noise floor and the shortest detectable ramp improve.
2. **Convective season 2027:** the real test. Gust fronts and microburst outflows will show whether the WSA and MBA levels and the confirmation rule hold.
3. **Validation** against METAR/SPECI windshear remarks, go-arounds (detected automatically), and pilot reports where available.
4. **Departure ramps:** the trial runs from 27 October (Section 8); check the stored climb-outs after one to two weeks — the effect of the acceleration on TAS − GS and the noise with few replies — before they may alert.
5. **Turboprop limits and type offsets:** review the factor 1.3 when more turboprop departures (and other types than the ATR 72) are stored; decide whether to remove the small per-type offsets (up to 1.4 kt) from the band winds.
6. **Space–time diagram** — *done 27 October* (Section 8); use it on the next fronts and convective days.
7. **Level review** after the first months, and an **EFHK low-level wind climatology** from the stored band profiles.
8. **BDS 4,5**: a validated decoding of the aircraft's own hazard flags would add the state of the on-board reactive system.

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